

EECE 4632

Hardware-Software Codesign for FPGA-Based Systems

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Today's Lecture:

- Projects
 - What you (hopefully) learned from this class
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- **Project Presentations during class on April 13 and 15. Finals slot on April 23.**
 - **Preliminary project report with code due April 14**
 - **Final project report with code due April 24**
 - **You may continue to work on your project up to the final project report deadline**

Project Presentations

- Your audience is the other students in the class
 - Imagine you are listening to someone else's presentation
 - What would you like to hear?
- 1. Describe your project.
- 2. What are your goals? What goes in hardware and what in software?
- 3. Present results/speedups, ...
- 4. How do you know that your project works? Validation. Test cases ...
- 5. Describe lessons learned. What would you do differently? What surprised you?
- 6. Future work: what will you do by the end of the semester?
What would you do if you had more time?

Q & A

Peer Project Evaluation

- Name one thing you liked about the project.
- Name one thing you think could be improved about the project.
- Name one thing you liked about the presentation.
- Name one thing you think could be improved about the presentation.
- Other comments?

Project Reports

- Similar to project presentation, but added detail:
 1. Describe your project.
 2. Describe your code.
 - Where did you get it? Where are the inputs from?
- What goes in hardware and what in software?
 1. Present results/speedups ...
 2. How did you test your code?
 3. Describe lessons learned. What would you do differently? What surprised you?
 4. Future work:
 1. What will you do for the rest of the term?
 2. What would you do if you had more time?
- 5. How did you make use of AI
- 6. Turn in code, scripts, instructions for running ...

Report advice

- The number of pages is NOT directly proportional to your grade!
 - Do not include code in your report. Summarize, write pseudo-code ...
 - Code should be handed in separately
 - Only include screen shots if they contain useful information. Put results in tables.
 - If your input data is huge upload it somewhere and submit a link.
- Do explore design tradeoffs
 - Hardware vs Software
 - Other design choices
 - HLS Design Space Exploration
- Best way to hand things in is via a project on GITHUB

Proper References

- ALWAYS REFERENCE YOUR SOURCES
- Where is your code from?
- Where are your input data from?
- It is fine to get code, images, etc. from somewhere else, but you need to give credit for where it came from.
- Here is a good source for how to reference things:
 - <https://www.york.ac.uk/students/studying/skills/integrity/referencing-styles/ieee/>
 - When you provide web sources, include date when you referenced them.
- Put references at the *end* of your report.

Final Project Grading

- Final project grade includes:
 - Final code and report grading (75 points)
 - Project update grades (10 points)
 - Presentation (8 pts)
 - Attendance at presentations (2 pts)
 - Final project grading:

Complexity	5
Hardware exploration	25
HW vs. software	10
Code runs	10
Quality of report	10
References	2
Subjective evaluation	18
Total	75

What is due on April 14

- A preliminary project report
- Code: Python, C, Jupyter Notebooks, and bit files so we can run the code
- Instructions on how to run your code and/or
 - set up an appointment to demo your code
- Inputs to run your code on or a link regarding where to find them
- Plan for the remainder of the semester

AXIS and your interfaces

```
typedef ap_axis<32,1,1,1> AXI_VAL
```

- 32 is the wordlength of the data you are streaming in

```
AXI_VAL output;  
output.data = acc;  
output.keep = tmp1.keep;  
output.strb = tmp1.strb;  
output.last = tmp1.last;  
output.dest = tmp1.dest;  
output.id = tmp1.id;  
output.user = tmp1.user;  
y.write(output)
```

“last” is a one bit signal that says the data transfer is finished. Pull it out of any loop calculating the result.

Calling your IP from a Jupyter Notebook

```
CONTROL_REGISTER = 0x0  
hls_ip.write(CONTROL_REGISTER, 0x01)
```

For more information check the AXI stream tutorial under files

4632: What you learned

- HW/SW codesign
 - What works well in software?
 - What works well in hardware?

Hardware Design is hard

- Hardware design is harder than software
- High Level Synthesis helps:
 - Design Space exploration is easier
 - Tools take care of the details
 - But ... still need to get the interfacing right
- Debugging:
 - Cannot just sprinkle print statements through out your design

Interfacing is Hard

- Getting data to/from hardware efficiently is hard
- Interfacing is done separately from HLS
- AXI is an ARM standard
 - Standards are good, but AXI is complex

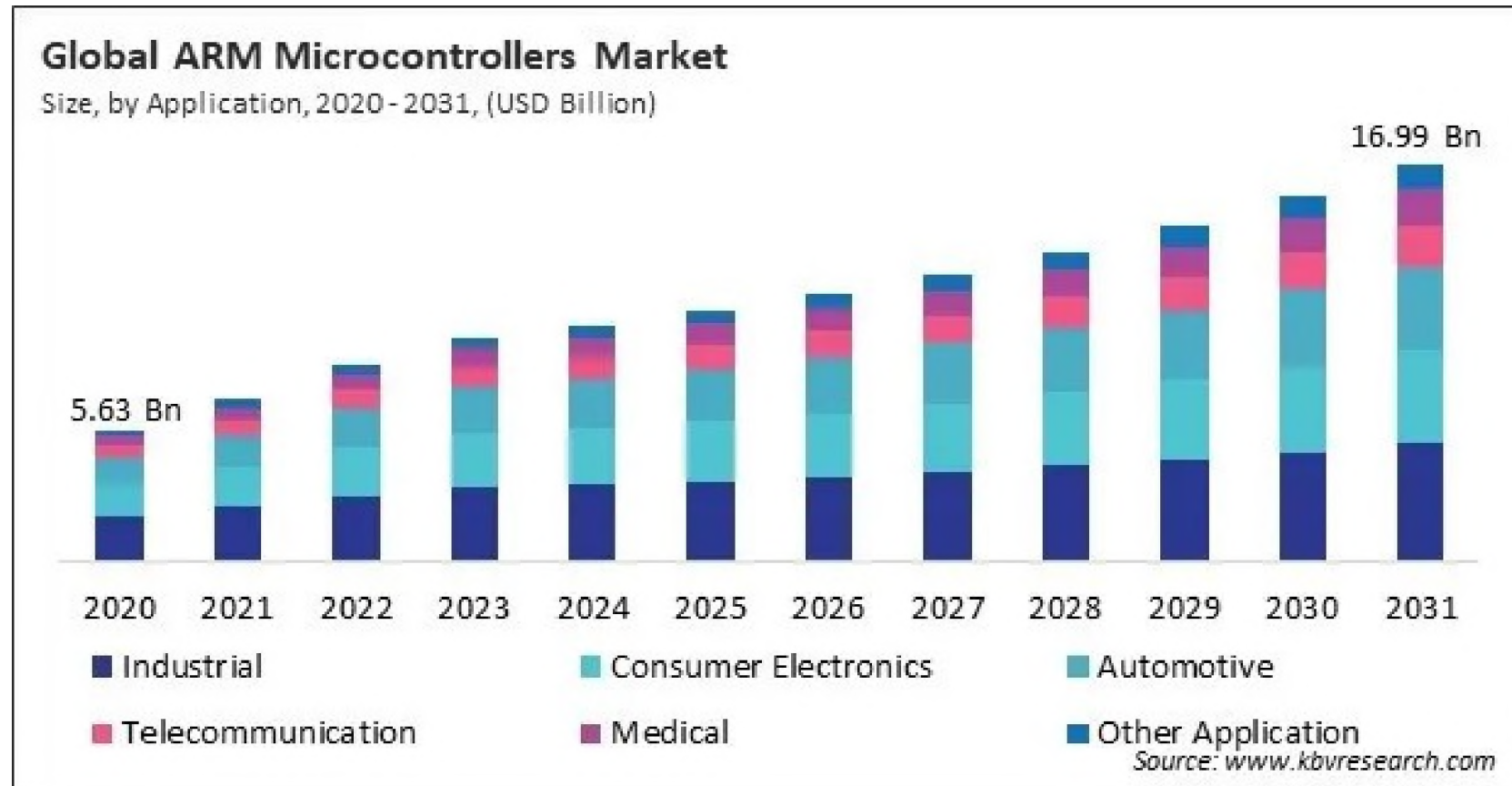
<https://docs.amd.com/v/u/en-US/ug1037-vivado-axi-reference-guide>

- Get interfaces right
 - Hardware and software can overlap computation

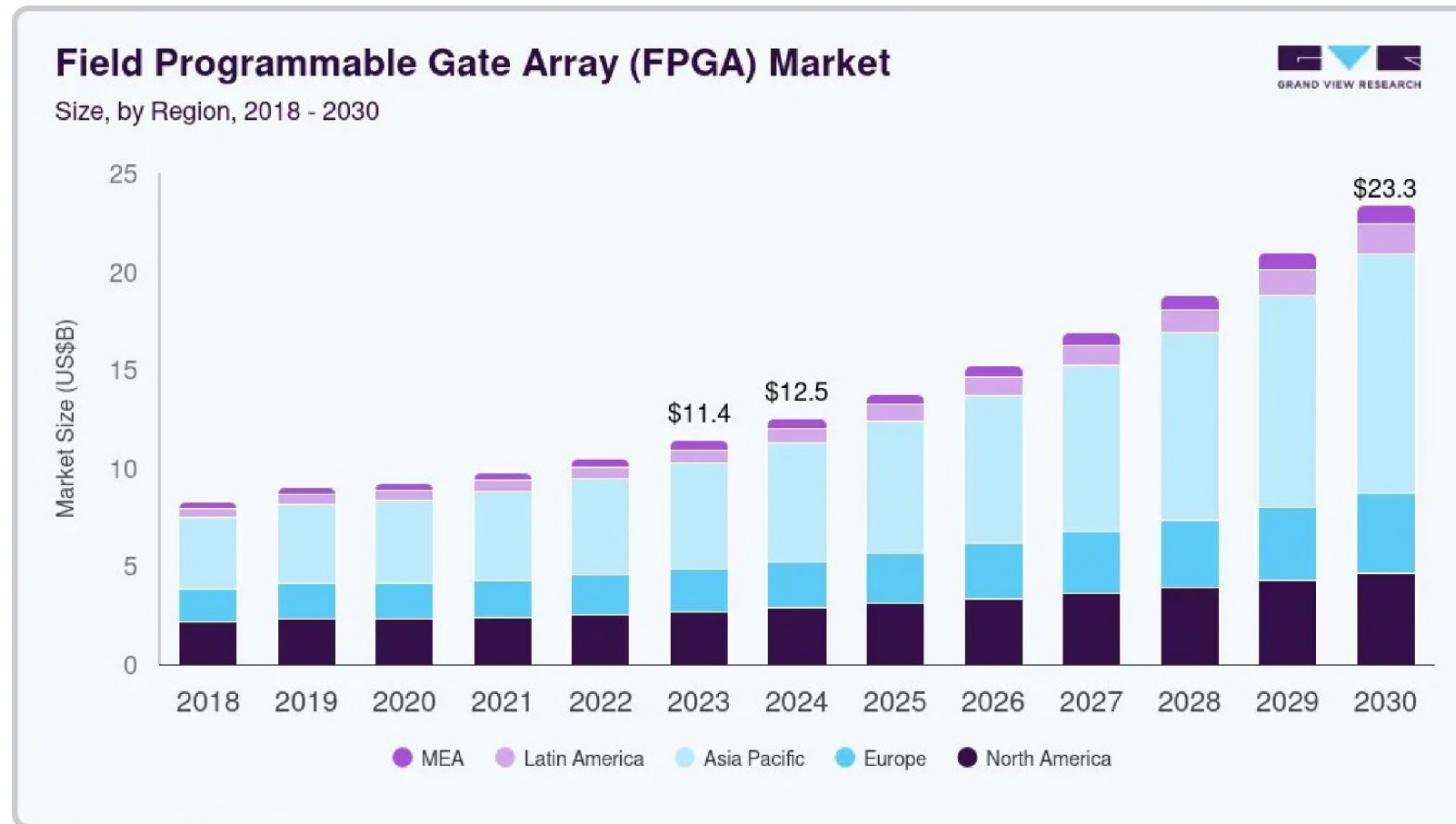
Data Representation Matters

- Different types and formats
- Numeric formats:
 - integer, fixed point, floating point
- Data representation affects:
 - complexity of operations
 - size of hardware
 - speed of hardware

<https://www.kbvresearch.com/arm-microcontrollers-market/>



FPGA Market Size



<https://www.grandviewresearch.com/industry-analysis/fpga-market>

<https://dataintel.com/report/soc-fpga-market>

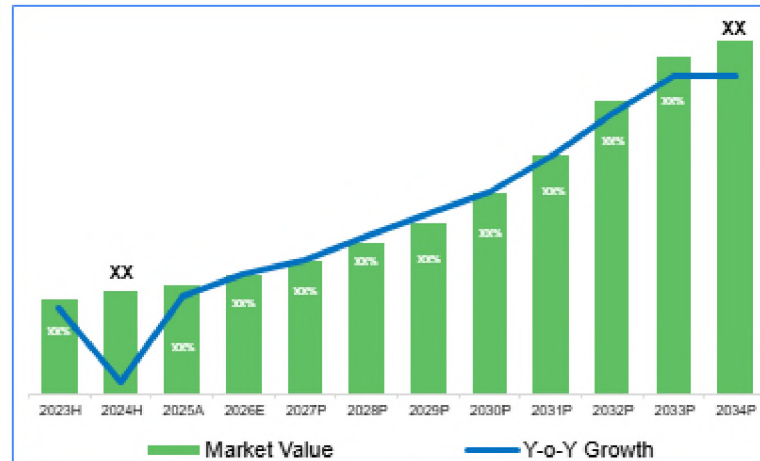


Fig XX: Global SoC FPGA Market Size & Forecast

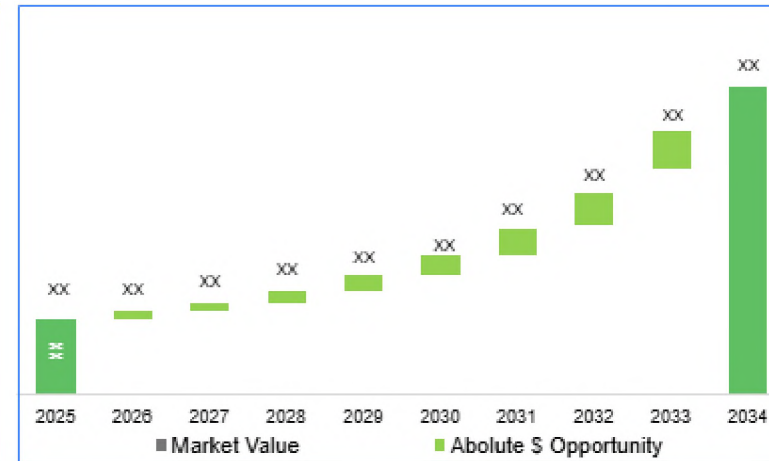
Global Outlook

2023-2034

Global SoC FPGA Market
Market Size (USD Million), Forecast and Y-o-Y Growth, 2023-2034



Global SoC FPGA Market
Market Absolute \$ Opportunity, in USD Million, 2025-2034



By Type

Low-End SoC FPGA
Mid-Range SoC FPGA
High-End SoC FPGA

By Technology

28nm
20nm
16nm
14nm

By Application

Automotive
Consumer Electronics
Industrial
Data Processing

By End-User

OEMs
ODMs
Others

Source: Dataintel Analysis

Email: sales@dataintel.com

Website: dataintel.com

Lots of challenges

- Zynq is an embedded system
- Embedded systems are growing in popularity
- So why are FPGAs and SoCs (FPGA + ARM) so popular?